

Elements of Position

IV. Meetings

A's spokes meet B's ratio. The docking has a closed form.

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2 is named by 1.

A note. Part I named the pair. Part II walked one street and split two mixes. This part lays a second street on the same midline and measures the meetings. Tables cited live under `kernels/ngon/runs/2026-09-24_n1-60/` and `measures/bridge/runs/2026-09-24/`.

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Named by one

1 names 2 by inverting the primary cut of Part I: $2 = 1/(1/2)$. On the pair $(x, 2 - x)$ the arithmetic mean is constantly 1, and equals the geometric mean only at $x = 1$.

The second street is not a new unit. It is the same unit asked the other way. After arrival it walks back toward the lock. Call the outward integer count street A and the inward golden walk street B, once, and then use the marks.

The common object is the line

The common object is not a circle. It is the $1/2$ line.

Every local diameter is cut at its midpoint. Street A's even split lives there. Street B's pair φ and $1/\varphi$ lives equally far from it. The cone of A, height the count, has slope $1/2$.

Freeze n : an n -gon of diameter n , a half-walk, a φ mark, an inward copy of radius $(n/2)/\varphi$. All of that is a section. Start the count and the same marks ride A outward and, after arrival, walk B the other way.

Marriage at five

Proposition 1.

$$2 \cos(\pi/5) = \varphi.$$

The n -gon kernel writes this as column `twocos` of `rings.csv`. On the 24 September run, at $n = 5$,

$$\text{twocos} = 1.618033988749895,$$

difference from φ equal to 0 at double precision.

The same five, asked of the walk by the vertex mix $h = \tan(2\pi/5)$: one elementary step per vertex, max error 4.44×10^{-16} , closing distance 3.4×10^{-17} , five unit sides equal to $2 \sin(\pi/5)$ with error 0.

That is the bridge row $n = 5$. A's own split at five is B's ratio, and the walk at the vertex mix is the pentagon.

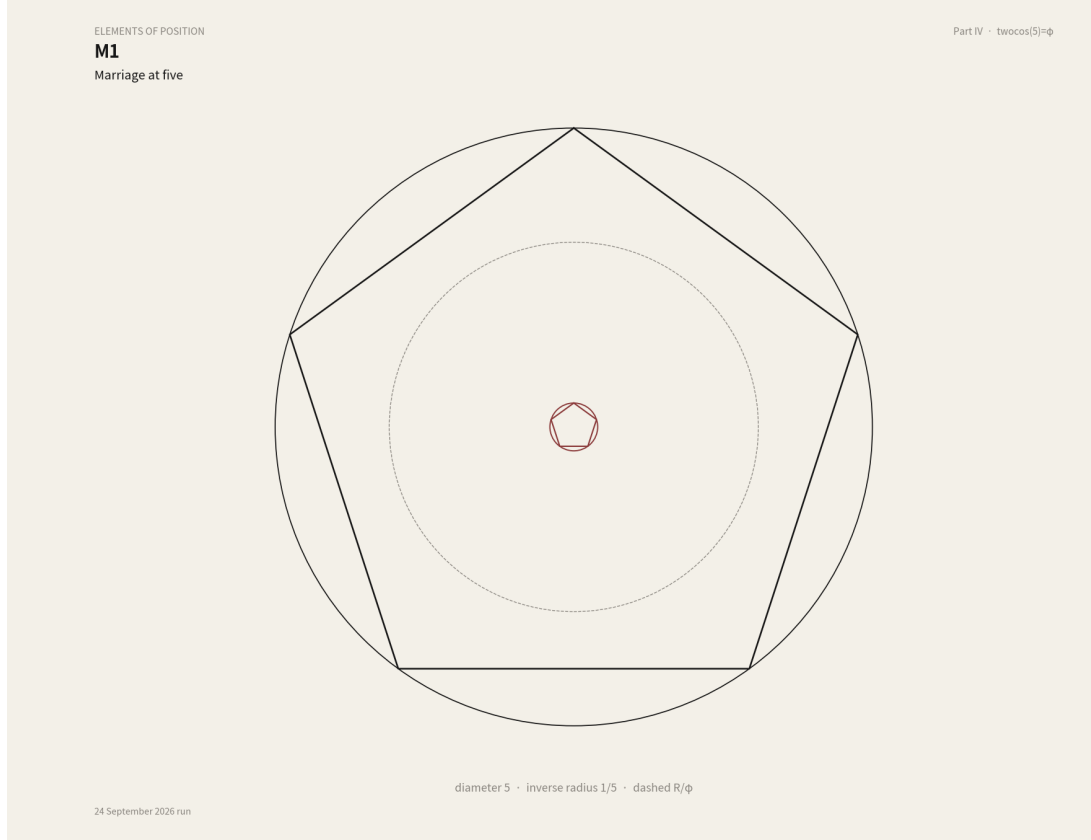


Figure 1: Sheet M1. Regular 5-gon of diameter 5. Dashed inward copy at R/φ . Accent is the inverse locus of radius $1/5$.

It does not spread to a generic later integer. It is not the half-turn mix. The half-turn mix at $n = 5$ is $h = \tan(\pi/5)$, one gap letter 5. The vertex mix at $n = 5$ writes gaps $\{2, 3\}$. Both are true. They are two kicks.

At the recorded grain $h = 0.002$, placing the same pentagon by walking rounded fractions of $2H$ steps on a circle of diameter 5 leaves a visible miss: max angle error 0.00145, max side error 0.00292. Spokes remain 2.5. That miss is Part III's residual, not a failed marriage.

Docking

Fibonacci docking is a different kind of meeting: approximate, after arrival, shrinking as the count runs.

Proposition 2. $F_k\varphi - F_{k-1} = -\psi^k$ exactly, where $\psi = (1 - \sqrt{5})/2$. The docking error is

$$\frac{1}{2}|F_k\varphi - F_{k-1}| = \frac{1}{2\varphi^k}.$$

Observation 1. At $k = 7$ ($N = F_7 = 13$), $1/(2\varphi^7) = 0.0172209$, matching the older printed “about 0.017.” At $k = 13$ ($N = 233$), $1/(2\varphi^{13}) = 0.0009597$, matching “about 0.00096.”

Cassini's identity is the case $M = RL$ of unimodularity on the Stern–Brocot tree. The identical mechanism with $\delta = 1 + \sqrt{2}$ gives a Pell analogue whose conjugate is smaller than $1/\varphi$, so that docking,

were it built, would converge strictly faster. It is not built here.

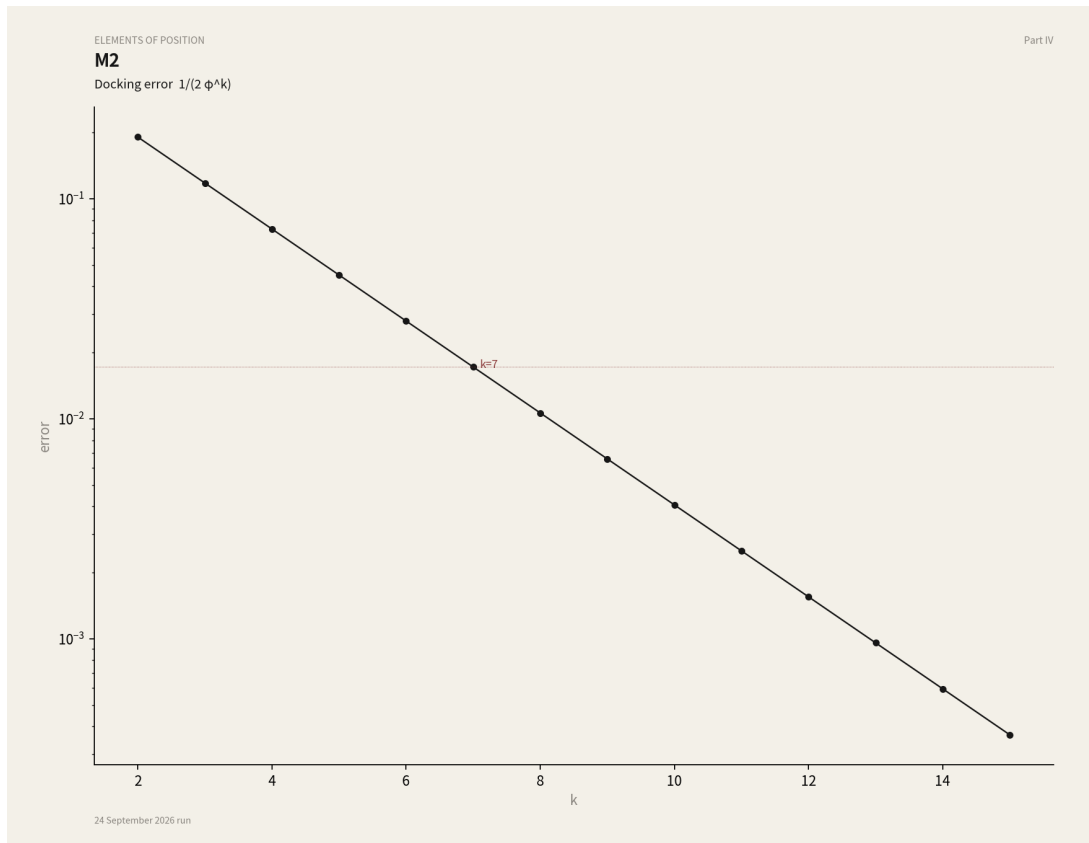


Figure 2: Sheet M2. Docking error $1/(2\phi^k)$. The mark at $k = 7$ is the old printed 0.017.

What stands

Forced: 2 named by 1; the $1/2$ line as the common object; marriage at five, two routes, machine agreement on the 24 September tables; docking error $1/(2\phi^k)$.

Not granted: marriage at a generic integer; the vertex mix spoken as if it wrote one gap letter n ; a numeric relation between γ and φ .

Where is 1? It is the lock both streets still share.